LAB 1 · PRE-LAB SETUP

Get your laptop ready

67-290 Storytelling with Data Visualization · CMU-Qatar

Do this before class on Wednesday 28 October · about 20–30 minutes

In Lab 1 you will build a live dashboard on your own laptop. Installing software on every laptop at once can eat the whole class, so please do these steps at home. None of it is hard; most of the time is waiting for downloads. You need your Mac or Windows laptop, an internet connection, and your CMU email (`yourID@andrew.cmu.edu`).

Your checklist

- ☐ 1 · Install VS Code
- ☐ 2 · Install Git
- ☐ 3 · Install Node.js
- ☐ 4 · GitHub account
- ☐ 5 · Final check

Bring to class

- ☐ Your laptop and charger
- ☐ Your GitHub username and password

Before you start: open a terminal

The terminal is a window where you type commands instead of clicking. You will use it in Steps 1, 2, 3 and 5.

- **Mac:** press `Cmd` + `Space` , type **Terminal**, press `Enter` .
- **Windows:** press the **Windows key**, type **PowerShell**, press `Enter` .

To run a command, type or paste it, then press `Enter` . Paste with `Cmd` + `V` on Mac or `Ctrl` + `V` on Windows.

After you install anything, close the terminal and open a new one. A terminal that was already open will not see the new tool.

1 Install VS Code

VS Code is the editor you will use to write your dashboard's files.

MAC

1. Go to **code.visualstudio.com** and click **Download for macOS**.
2. In your **Downloads** folder, drag **Visual Studio Code** into **Applications**. (If you see a `.zip` file, double-click it first.)
3. Open VS Code from Applications. If macOS asks whether you are sure, click **Open**.
4. In VS Code, press `Cmd` + `Shift` + `P` , type **shell command**, and click **Shell Command: Install 'code' command in PATH**. This lets you open VS Code from the terminal, which you will do in the lab.

WINDOWS

1. Go to **code.visualstudio.com** and click **Download for Windows**.
2. Run the installer and accept the defaults. Keep **Add to PATH** checked.

Check: open a new terminal and run `code --version` . The first line it prints is a version number.

2 Install Git

Git keeps a history of your work, like an undo button for a whole folder.

MAC

1. In Terminal, run:

   ```
   git --version
   ```

2. If it prints something like `git version 2.x.x`, Git is already installed. Go to Step 3.
3. If a window asks to install the **command line developer tools**, click **Install** and agree. This can take several minutes.
4. When it finishes, run `git --version` again.

No window appeared? Run `xcode-select --install`.

WINDOWS

1. Go to **git-scm.com/downloads**, click **Windows**, and download the installer.
2. Run it and accept every default, with one exception: on the screen about the **default branch name**, choose **Override the default branch name for new repositories** and keep `main`.
3. Close PowerShell, open a new one, and run:

   ```
   git --version
   ```

Check: `git --version` prints `git version` followed by a number.

3 Install Node.js

Node.js runs the JavaScript behind your dashboard. Already have it? Run `node -v`. If it prints `v22` or higher, skip to Step 4.

MAC

Run these four commands one at a time. They install **nvm**, a small tool that installs Node, and then Node 22.

```
touch ~/.zshrc
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.3/install.sh | bash
. "$HOME/.nvm/nvm.sh"
nvm install 22
```

The first line makes sure Terminal still finds Node the next time you open it.

WINDOWS

1. Download the Node.js 22 installer: nodejs.org/dist/v22.23.4/node-v22.23.4-x64.msi
2. Run it and accept the defaults.
3. Close PowerShell and open a new one.

Check: in a new terminal, `node -v` prints `v22` or higher, and `npm -v` prints a version number.

4 Set up your GitHub account

GitHub is where your work goes online, and where the TA grades it. Find your situation and follow only that part.

A. You don't have a GitHub account.

1. Go to **github.com** and click **Sign up**.
2. Use your CMU email (`yourID@andrew.cmu.edu`), create a password, and choose a username.
3. Enter the code GitHub sends to your CMU inbox. You can skip the welcome survey.

B. You have an account, but with a personal email. Don't make a second account; that splits your history across two profiles. Add your CMU email instead:

1. Sign in, click your profile photo (top right), then **Settings** → **Emails**.
2. Add `yourID@andrew.cmu.edu` , then click the verification link GitHub sends to your CMU inbox.

C. You have an account with your CMU email. Nothing to do.

Pick a username you are happy to keep. The TA uses it to grade your labs, and it is the front page of your portfolio for years afterwards.

Check: on GitHub, **Settings → Emails** shows your CMU email as **Verified**, and you can sign in with your username and password.

5 Final check

Open a **new** terminal and run these four commands:

```
code --version
git --version
node -v
npm -v
```

If all four print a version number and you can sign in to github.com, **you are ready for Lab 1.**

Something not working?

- `command not found` (Mac) or `is not recognized` (Windows) right after installing: close the terminal completely and open a new one. If that doesn't help, restart your laptop.
- Mac, `code: command not found` : repeat Step 1, item 4, then open a new terminal.
- Mac, `nvm: command not found` or `node: command not found` in a new terminal: run `touch ~/.zshrc` , then the `curl` line from Step 3 again, then open a new terminal.
- Windows, `npm` says **running scripts is disabled on this system**: run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` , type `Y` , then open a new PowerShell.
- Still stuck? Email the TA at `peidid@andrew.cmu.edu` with a screenshot of the error, or bring your laptop to class and tell us at the start. Part 1 of the lab begins with these tools.